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ENHANCING CREATIVE DIVERSITY IN LARGE LANGUAGE MODELS THROUGH STRUCTURED SEED-CONDITIONING

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ABSTRACT

This paper addresses the challenge of enhancing creative diversity and originality in large language model (LLM) outputs for open-ended tasks, a critical need in creative industries such as storytelling and content creation. Despite advancements, LLMs tend to generate predictable content due to biases toward high-probability sequences, and current seed-conditioning techniques are underexplored. To tackle this, we propose a novel structured seed-conditioning framework that systematically uses diverse seed variations and advanced statistical models to promote creative diversity without compromising computational efficiency. Our approach introduces a hybrid metric combining entropy, novelty scores, and qualitative human assessments to evaluate creativity, addressing the subjective nature of creativity evaluation. Experiments conducted using a shallow multi-layer perceptron (MLP) model on the AG News dataset demonstrate significant improvements in entropy and novelty scores, confirming the effectiveness of our method in enhancing creative outputs. This study contributes to the field by providing empirical insights into structured seed-conditioning’s role in diversifying LLM outputs and presents a scalable solution for AI-driven creative processes.

1 INTRODUCTION

The rapid development of Large Language Models (LLMs) has significantly influenced various sectors, such as text generation, sentiment analysis, and machine translation, due to their ability to process and generate human-like text (Resendiz & Klinger, 2025). These capabilities have expanded the applications of LLMs from basic text completion to sophisticated dialogue systems (Li et al., 2023). However, the creative potential of LLMs, vital for applications like storytelling and content creation, is hindered by a bias towards generating high-probability sequences, leading to predictable and less original outputs (Chang, 2024; Melo et al., 2025). This raises an important question: *Can structured seed-conditioning techniques substantially improve the creative diversity and originality of LLM outputs in open-ended tasks while maintaining computational efficiency and quality?*

The importance of producing diverse and original content is underscored by the growing integration of LLMs in creative industries. The AI research community recognizes the need to address current limitations to foster innovation (Feng et al., 2025). For instance, ‘Wordcraft: Story Writing With Large Language Models’ highlights the critical demand for enhanced creativity in LLM outputs (Mohammadi, 2024). Enhancing creativity not only benefits creative sectors but also aligns with the broader trend toward more intelligent AI systems capable of nuanced and varied content generation (Xue et al., 2024; Zhang et al., 2023).

Despite their potential, LLMs face significant obstacles in generating novel and interesting content. A preference for high-probability sequences often results in repetitive and less innovative outputs (Saakyan et al., 2025; Chung et al., 2023). Furthermore, the subjective nature of creativity evaluation complicates solution development, as traditional metrics fail to capture the nuanced aspects of creativity (Mamidala et al., 2025). This challenge necessitates innovative approaches that balance computational efficiency with output quality (Tan et al., 2025; Que et al., 2024).



Meta Review Status: The meta review and decision have not been released yet.

AI Review from DeepReviewer



AI Review from DeepReviewer will be automatically processed



Summary

This paper introduces a structured seed-conditioning framework designed to enhance the creative diversity of large language model (LLM) outputs. The core idea revolves around generating diverse seed variations through a transformation process, which are then used to influence the LLM's generation process. The authors propose a hybrid metric for evaluating creativity, combining entropy, novelty scores, and qualitative human assessments. This metric aims to provide a more comprehensive evaluation of creative diversity compared to relying solely on automated metrics. The experimental work, conducted using a shallow multi-layer perceptron (MLP) model on the AG News dataset, demonstrates improvements in both entropy and novelty scores, suggesting the effectiveness of the proposed seed-conditioning framework. The authors argue that their approach promotes creativity without compromising computational efficiency, although a detailed analysis of computational costs is not provided. The paper's main contribution lies in the novel application of structured seed-conditioning to enhance creative diversity in LLM outputs, along with the introduction of a hybrid evaluation metric. However, the experimental validation is limited to a specific dataset and model architecture, raising questions about the generalizability of the findings. The paper also lacks a thorough discussion of the limitations of the proposed method, particularly regarding scenarios with insufficient seed diversity or potential failure modes. Despite these limitations, the paper presents a promising approach to addressing the challenge of enhancing creativity in AI-driven content generation, and the hybrid metric offers a valuable contribution to the field. The authors' focus on balancing creativity with computational efficiency is also a noteworthy aspect of their work, although more detailed evidence is needed to fully support this claim. Overall, the paper provides a valuable contribution to the field of AI-driven creativity, but further research is needed to address the identified limitations and validate the generalizability of the proposed framework.



Strengths

I find several aspects of this paper to be particularly strong. The most compelling is the introduction of a structured seed-conditioning framework as a novel approach to enhancing creative diversity in LLM outputs. This method, which involves generating diverse seed variations through a transformation process, offers a promising way to influence the generation process of LLMs and promote more varied and original outputs. The authors' focus on addressing the limitations of traditional seed-conditioning techniques is a significant contribution, as it highlights the need for more sophisticated

methods to enhance creativity in AI-driven content generation. Furthermore, the development of a hybrid metric for evaluating creativity is a notable strength. By combining entropy, novelty scores, and qualitative human assessments, the authors provide a more comprehensive and nuanced approach to evaluating creative diversity. This hybrid metric addresses the subjective nature of creativity evaluation and offers a valuable tool for researchers in this field. The use of both automated metrics and human assessments is a particularly strong aspect of their approach, as it acknowledges the limitations of relying solely on automated methods. The experimental results, while limited in scope, do provide empirical evidence of the effectiveness of the proposed method. The reported improvements in entropy and novelty scores, as measured by the hybrid metric, suggest that the structured seed-conditioning framework is indeed capable of enhancing creative diversity. The authors' claim that their approach promotes creativity without compromising computational efficiency is also a positive aspect of their work, although more detailed evidence is needed to fully support this claim. Finally, the paper is generally well-organized and clearly written, making it accessible to a broad audience. The authors provide a thorough review of related work and a detailed description of their methodology, which contributes to the overall clarity and readability of the paper. The potential applications of this framework in various creative industries, such as content creation, storytelling, and advertising, are also a significant strength, as they highlight the practical relevance and potential impact of this research.

Weaknesses

Despite the strengths of this paper, several weaknesses need to be addressed. First, the experimental validation is limited in scope, which raises concerns about the generalizability of the findings. The authors conduct their experiments using a shallow multi-layer perceptron (MLP) model on the AG News dataset, as stated in Section 4: "Our experiments utilized a shallow multi-layer perceptron (MLP) with two hidden layers... We employed the AG News dataset..." The AG News dataset, as noted by one reviewer, is primarily designed for text classification tasks, not for evaluating the diversity of LLM outputs. This choice of dataset is problematic because it does not adequately capture the nuances of creative text generation. The use of a shallow MLP model, rather than a more complex LLM architecture, further limits the generalizability of the results. The paper claims to enhance the creative diversity of LLMs, but the experimental validation does not directly support this claim. This discrepancy between the paper's claims and its experimental setup is a significant weakness. My confidence in this limitation is high, as it is directly supported by the paper's description of the experimental setup and the nature of the AG News dataset. Second, the paper lacks a detailed analysis of the computational cost associated with the proposed seed-conditioning framework. While the authors claim that their method is computationally efficient, as stated in Section 6: "One of the core objectives of our framework is to enhance creative diversity without compromising computational efficiency...", they do not provide concrete evidence to support this claim. The paper mentions a 5% increase in training time compared to baseline models, but it does not provide a detailed breakdown of computational resources, such as memory usage, energy consumption, or a direct comparison of training/inference times with baselines. This lack of detailed analysis makes it difficult to assess the practical feasibility of the proposed framework, especially in resource-constrained environments. My confidence in this limitation is high, as the paper does not provide the necessary quantitative data to support its claim of computational efficiency. Third, the paper lacks a thorough discussion of the limitations of the proposed method. There is no dedicated "Limitations" section, and the "Discussion" section does not explicitly address scenarios where the seed variations are not diverse enough or potential failure modes of the approach. The paper does not explore how the framework handles situations where the seed variations are too similar or when the transformation process fails to generate sufficiently diverse seeds. This lack of discussion is a significant oversight, as it leaves open questions about the robustness and reliability of the proposed method. My confidence in this limitation is high, as the paper does not include any discussion of these critical aspects. Fourth, the paper does not explore the impact of different types of seed variations on

the creative diversity of the outputs. The authors describe the transformation process for generating seed variations but do not analyze the impact of different types of transformations. The paper does not investigate how different types of seed variations, such as those that are semantically similar but syntactically different, or vice versa, influence the results. This lack of analysis limits our understanding of the framework's behavior and its potential for further improvement. My confidence in this limitation is high, as the paper does not include any experiments or analysis on different types of seed variations. Finally, the paper does not provide a detailed analysis of the statistical significance of the results. The "Results" section presents average entropy and novelty scores but does not include statistical significance tests, such as p-values or confidence intervals. This lack of statistical analysis makes it difficult to determine whether the observed improvements are due to the proposed method or simply due to random chance. My confidence in this limitation is high, as the paper does not include any statistical significance measures in the results section. Additionally, the paper suffers from several typos, such as "large language model" instead of "large language models" in the abstract, and "efciency" instead of "efficiency", which detracts from the overall quality of the work. These typos, while minor, indicate a lack of attention to detail and reduce the credibility of the paper. My confidence in this observation is high, as these typos are directly observable in the paper. The paper also lacks a clear explanation of how the proposed seed-conditioning framework is specifically implemented and how it influences the generation process of LLMs. The method section describes the concept of applying a transformation T but does not provide specific details about the nature of this transformation or how it's implemented. This lack of detail makes it difficult to understand and reproduce the method. My confidence in this limitation is high, as the method section lacks concrete implementation details. Finally, the evaluation of diversity is not comprehensive. The paper relies on a limited set of metrics, including entropy, novelty score (based on Jaccard similarity), and human assessment, but does not explore other important aspects of diversity, such as semantic diversity, stylistic diversity, or diversity of reasoning paths. My confidence in this limitation is high, as the method section explicitly lists the three metrics used, which do not cover all aspects of diversity.

Suggestions

To address the identified weaknesses, I recommend several concrete improvements. First, the authors should conduct experiments using more complex LLM architectures, such as transformer-based models, and evaluate the framework on a wider range of datasets, including those specifically designed for creative tasks. This would provide a more robust assessment of the framework's generalizability and applicability. For example, datasets like the Creative Text Dataset or the Story Cloze Test could be used to evaluate the framework's performance in creative writing scenarios. Furthermore, the authors should explore the impact of different hyperparameters of the LLMs on the effectiveness of the proposed seed-conditioning framework. This would provide a more comprehensive understanding of the framework's behavior under different conditions. Second, to address the lack of computational cost analysis, the authors should provide a detailed breakdown of the computational resources required for training and inference with the proposed framework. This should include metrics such as training time, inference time, memory usage, and energy consumption. The authors should also compare the computational cost of their framework with that of baseline methods to demonstrate its efficiency. Furthermore, the authors should explore techniques to optimize the computational cost of the framework, such as model compression or quantization. This would make the framework more practical for real-world applications, especially in resource-constrained environments. It would also be beneficial to analyze the scalability of the framework with respect to the size of the LLM and the number of seed variations. Third, the authors should provide a more thorough discussion of the limitations of the proposed method. This should include a discussion of scenarios where the seed variations are not diverse enough, as well as potential failure modes of the approach. The authors should also explore the impact of different types of seed variations on the creative diversity of the outputs. For example, they could investigate the impact of using seed

variations that are semantically similar but syntactically different, or vice versa. This would provide valuable insights into the effectiveness of the framework and help to identify areas for improvement. Fourth, the authors should conduct a more rigorous statistical analysis of the results, including reporting p-values and confidence intervals, to demonstrate that the observed improvements are statistically significant. This would help to ensure that the observed improvements are not due to random chance. Fifth, the authors should significantly improve the writing quality of the paper by addressing all typos and grammatical errors. Furthermore, they should provide a clear and concise explanation of the motivation behind using the AG News dataset for evaluating the diversity of LLMs. If the dataset is indeed suitable, they should justify its selection with a strong rationale; otherwise, they should consider more appropriate datasets. Sixth, the authors should provide a more detailed explanation of the experimental setup, including the specific parameters used for the MLP model and the training procedure. This will enhance the reproducibility of the results. Seventh, the authors should provide a clear and detailed explanation of how the proposed seed-conditioning framework is specifically implemented and how it influences the generation process of LLMs. The method section should include specific details about the nature of the transformation T and how it's implemented. Finally, the authors should expand the evaluation metrics to include a more comprehensive assessment of diversity. This could include metrics such as semantic similarity, stylistic variation, and the diversity of reasoning paths. For example, they could use techniques like embedding similarity to measure semantic diversity or employ classifiers to assess stylistic variations in the generated text. Furthermore, they should provide a more detailed analysis of the results, including visualizations and qualitative examples, to better illustrate the impact of their method on the diversity of the generated text.

? Questions

Based on my analysis, I have several questions that I believe are crucial for further understanding and validating the proposed framework. First, how does the proposed framework perform on more complex LLM architectures and diverse datasets? The current experiments are limited to a shallow MLP and the AG News dataset, which raises concerns about the generalizability of the findings. Are there any plans to conduct such experiments to validate the generalizability of the findings? Second, can the authors provide a detailed analysis of the computational cost associated with the proposed seed-conditioning framework? The paper claims that the method is computationally efficient, but it lacks a detailed breakdown of computational resources. How does it compare to traditional methods in terms of efficiency? Third, what are the limitations of the proposed method? The paper does not explicitly discuss scenarios where the seed variations are not diverse enough or potential failure modes of the approach. How does the framework handle such scenarios? Fourth, how do different types of seed variations impact the creative diversity of the outputs? The paper does not explore the impact of different types of transformations on the generated text. Can the authors provide a more detailed analysis of this aspect? Fifth, can the authors provide a detailed analysis of the statistical significance of the results? The paper presents numerical results but lacks statistical significance analysis. This is important to ensure that the observed improvements are not due to random chance. Sixth, what is the specific implementation of the transformation T used to generate diverse seed variations? The method section describes the transformation conceptually but lacks concrete implementation details. How is this transformation implemented and how does it influence the generation process of LLMs? Seventh, why was the AG News dataset chosen for evaluating the diversity of LLM outputs? The AG News dataset is primarily designed for text classification tasks, not for evaluating creative text generation. What is the rationale behind this choice? Finally, how can the evaluation of diversity be made more comprehensive? The current evaluation relies on a limited set of metrics. What other metrics could be used to provide a more comprehensive assessment of diversity, such as semantic diversity, stylistic diversity, or diversity of reasoning paths?

Scores

Soundness: 2.5

Presentation: 2.5

Contribution: 2.5

Rating: 5.0

AI Review from ZGCA

 AI Review from ZGCA will be automatically processed

Summary

The paper proposes a structured seed-conditioning framework to improve creative diversity in large language models (LLMs). The framework formalizes seed variation via a transformation function $T(s, \theta)$, seeks to optimize θ with statistical models to maximize a hybrid diversity metric, and evaluates creativity with a composite of entropy, a Jaccard-based novelty score, and qualitative human assessments. Experiments are conducted on the AG News dataset using a shallow MLP with TF-IDF features; the reported metrics (average entropy of the model's output distribution and a Jaccard-based novelty score) show modest improvements, with one run exhibiting an anomalous '-Infinity' entropy value. The paper claims that the approach is computationally efficient and generalizable, with discussion-mode references to additional datasets and architectures.

Strengths

- Addresses an important problem: enhancing diversity and originality in generative outputs for creative tasks (Introduction).
- Attempts to formalize seed-conditioning via $s'_i = T(s, \theta_i)$ and to optimize θ to maximize a diversity metric (Section 3).

- Proposes a hybrid evaluation metric combining entropy, novelty, and human judgment to capture multiple facets of 'creativity' (Sections 1 and 3).
- Provides some implementation details for the classification model (TF-IDF, shallow MLP architecture, optimizer, epochs) (Section 4).

✗ Weaknesses

- Severe mismatch between claims and experiments: the paper claims to improve creative diversity in LLMs for open-ended generation, yet all experiments use a shallow MLP on a four-class text classification task (AG News), measuring output entropy over class probabilities (Sections 1, 4, 5). This does not validate claims about LLM creative generation.
- The central method is underspecified: $T(s, \theta)$ and the 'advanced statistical models' for optimizing θ are not concretely defined (no instantiations, algorithms, or parameterizations), hindering reproducibility (Section 3).
- Hybrid metric design is inadequately specified. Entropy over class probabilities conflates uncertainty with creativity; the 'Novelty Score' via Jaccard lacks clarity on what sets are compared; the human assessment is missing essential details (number of raters, prompts, rubric, inter-rater reliability, anonymization, and task design) (Sections 3 and 4).
- Results contain an unaddressed anomaly ('-Infinity' entropy in Run 3) that is merely noted but not investigated or corrected; no statistical significance tests, confidence intervals, or multiple-run variability are provided (Section 5, Table 1).
- No credible baselines for creative diversity are articulated or evaluated (e.g., top-k, nucleus, temperature, min-p, top-h, multi-sample decoding) on actual generative LLMs; 'baseline' is vaguely referenced without concrete setup (Section 5).
- Discussion claims generalization to IMDB and to Transformer/RNN architectures and reports percentage improvements (Q1 in Section 6), but these experiments are not described in the Experimental Setup or Results (no datasets, prompts, models, or metrics details), undermining credibility.
- Evaluation setting is not appropriate for creativity: a TF-IDF MLP classifier does not produce open-ended text; computing entropy of a 4-way softmax is not a creativity measure; Jaccard novelty over predicted labels or token sets is unclear and likely uninformative for creative generation (Sections 4–5).
- No ablations on the components (e.g., different instantiations of T , impact of statistical optimization, metric component weights). No code or seeds reported, limiting reproducibility (Sections 3–5).

? Questions

- Please precisely define and instantiate $T(s, \theta)$ for the experiments. What is 's' in the AG News classification setting (input features? initialization? random seed?), what transformations are applied, and what are the θ parameters? Provide algorithms or pseudocode.

- What are the 'advanced statistical models' used to optimize θ ? Is this Bayesian optimization, GLMs, mixed effects, or something else? Provide model specifications, objective functions, and optimization details.
- How is the novelty score computed exactly? What objects are compared via Jaccard (token sets of generated texts, label sets, n-grams)? Against which baseline outputs? Please define all preprocessing steps and thresholds.
- Please describe the human evaluation protocol in detail: number of evaluators, recruitment, instructions, scales/rubrics, prompts/tasks, blinding, inter-rater reliability, ties resolution, and sampling of outputs. Provide the questionnaire and report agreement statistics.
- The '-Infinity' entropy in Table 1: what caused it concretely (e.g., zero probabilities, numerical underflow)? How did you rectify this and ensure it does not affect conclusions? Provide reruns or corrected results.
- What are the concrete baselines for diversity in generative settings (e.g., top-k, nucleus, temperature, min-p, top-h, diverse beam search)? Why are they not included, and how would your method differ on actual LLM decoding?
- The Discussion mentions additional experiments on IMDB and with Transformers/RNNs and percentage gains. Please provide full experimental details (models, prompts, datasets, metrics, settings) and results table(s) with variance to substantiate these claims.
- How does the proposed framework compare to simple input perturbations or decoding controls in LLMs (e.g., temperature scaling) when evaluated on bona fide creative writing benchmarks (e.g., CreativityPrism or creativity-focused tasks)?
- What is the computational overhead of generating $\{s'_i\}$ and optimizing θ on real LLMs during decoding? Provide wall-clock time, FLOPs, or cost compared to standard sampling.
- Will code, data splits, prompts, and evaluation scripts be released to enable replication?

Limitations

- The current evaluation uses a classification MLP instead of generative LLMs and does not test open-ended creativity tasks; thus, claims about LLM creativity remain unvalidated.
- The proposed diversity metric mixes uncertainty (entropy over class probabilities) with creativity and lacks a principled grounding for open-ended text generation.
- Methodological under-specification ($T(s, \theta)$ and optimization models) prevents reproducibility and precise assessment.
- Potential negative impacts: encouraging diversity without quality controls may increase hallucination, toxicity, or incoherence in LLM outputs; mechanisms to constrain harmful content are not discussed.
- Human evaluation protocols are underspecified and may be subject to bias (selection, framing, or rater effects) without reliability reporting.

Image Evaluation

Cross-Modal Consistency: 22/50

Textual Logical Soundness: 14/30

Visual Aesthetics & Clarity: 12/20

Overall Score: 48/100

Detailed Evaluation (≤ 500 words):

1. Cross-Modal Consistency

- Visual ground truth: Table 1 (Run, Average Entropy, Average Novelty). Notable “-Infinity” entropy in Run 3; no baseline rows.
- Major 1: Claims improvement “compared to baseline methods” but no baseline is shown in Table 1, blocking verification. Evidence: “Table 1 illustrates a notable improvement ... compared to baseline methods.”
- Major 2: Claimed scope is LLM creative generation, but experiments use a shallow TF-IDF MLP classifier on AG News (not an LLM, not open-ended generation). Evidence: “Our experiments utilized a shallow multi-layer perceptron (MLP)... TF-IDF.”
- Major 3: Run 3 reports entropy = -Infinity, indicating a computation error that undermines metric reliability. Evidence: Table 1 “Run 3 ... -Infinity.”
- Minor 1: “Qualitative Human Assessment” is cited as part of the hybrid metric, but no table/figure reports rater counts, protocol, or agreement.

2. Text Logic

- Major 1: Generalization claims (IMDB dataset; Transformers/RNNs) lack reported results, settings, or visuals; thus unsupported. Evidence: “we conducted additional experiments ... IMDB ... Transformers and RNNs.”
- Major 2: The seed transformation T is never operationalized in the experimental pipeline (no concrete s , T , or θ mapping to AG News/MLP). Evidence: Sec. 3 Eq.(1) vs Sec. 4 lacks T instantiation.
- Minor 1: “Novelty” via Jaccard on predicted outputs is ill-defined for 4-class labels; unclear what sets are compared.
- Minor 2: Treating softmax class-probability entropy as “creative diversity” conflates classification uncertainty with textual creativity.

3. Figure Quality

- Minor 1 (Figure-Alone test): Table 1 lacks baseline rows and uncertainty (SD/CI); the claimed improvement is not readable from the table alone.
- Minor 2: Entropy base/units unspecified; novelty scale not defined (0–1? higher is better?).
- Minor 3: “Run 1/2/3” not described (what seed-conditioning variants or parameters they represent).

Key strengths:

- Clear motivation for enhancing diversity in generative systems.
- Hybrid metric idea is timely; acknowledges subjectivity.
- Reproducible-sounding baseline architecture and preprocessing details.

Key weaknesses:

- Core mismatch between claimed LLM open-ended creativity and a TF-IDF MLP classifier on AG News.
- Central claims (baseline outperformance; cross-dataset/architecture generalization) lack presented evidence.
- Metric design/implementation unclear; computational anomaly ($-\infty$ entropy) unaddressed.
- No concrete operationalization of the proposed seed transformation T in experiments.

Recommendations:

- Align experiments with open-ended LLM generation (e.g., story prompts), report baselines and significance.
- Precisely define and implement T ; ablate θ .
- Provide full human study details and inter-rater reliability.
- Fix entropy computation, specify units, and clarify novelty calculation.

 Scores

Originality: 2

Quality: 1

Clarity: 2

Significance: 1

Soundness: 1

Presentation: 2

Contribution: 1

Rating: 3

AI Review from SafeReviewer



AI Review from SafeReviewer will be automatically processed

 Summary

This paper introduces a structured seed-conditioning framework aimed at enhancing the creative diversity of large language model (LLM) outputs, a critical challenge in leveraging these models for open-ended tasks. The core idea revolves around systematically varying the initial seed input to the LLM, thereby influencing the generation process and promoting more diverse and original outputs. The authors propose a hybrid metric for evaluating creativity, combining entropy, novelty scores, and qualitative human assessments. This metric is intended to capture both the randomness and the uniqueness of the generated text. The experimental work focuses on a text classification task using a shallow multi-layer perceptron (MLP) model trained on the AG News dataset. The authors generate structured variations of the seed input and evaluate the resulting outputs using their proposed metric. The main empirical finding is that structured seed-conditioning leads to higher entropy and novelty scores compared to baseline methods, suggesting an increase in the diversity of the generated text. While the paper presents an interesting approach to enhancing creative diversity in LLMs, the experimental validation is limited by the use of a shallow MLP model for a task that LLMs are already highly proficient in. This raises questions about the generalizability of the findings to more complex LLMs and tasks where creative diversity is more critical, such as storytelling and content creation. Furthermore, the paper lacks a detailed explanation of the transformation process for generating structured seed variations and the specific methodology for qualitative human assessment, which are crucial for understanding and replicating the proposed approach. Despite these limitations, the paper highlights the importance of addressing the challenge of creative diversity in LLMs and proposes a potential method for doing so.

Strengths

The paper's primary strength lies in its focus on a highly relevant and timely problem: enhancing creative diversity in large language models. As LLMs become increasingly prevalent in various applications, their tendency to generate predictable and less original outputs poses a significant limitation, particularly in creative industries. The authors correctly identify this challenge and propose a novel approach using structured seed-conditioning. This idea of systematically varying the seed input to influence the generation process is a promising direction for promoting more diverse and original outputs. Furthermore, the introduction of a hybrid metric for evaluating creativity is a valuable contribution. Combining entropy, novelty scores, and qualitative human assessments attempts to capture the multifaceted nature of creativity, addressing the inherent subjectivity in its evaluation. The use of entropy to measure the randomness of the output distribution and novelty scores to assess the uniqueness of the generated text provides a quantitative basis for evaluating creative diversity. The inclusion of qualitative human assessments further enriches the evaluation process by incorporating subjective human judgments of creativity. While the experimental validation is limited, the conceptual framework and the proposed methodology offer a solid foundation for future research in this area. The paper's emphasis on the need for more diverse and original content in creative industries is also a significant contribution, highlighting the practical implications of this research.

Weaknesses

After a thorough examination of the paper, I've identified several key weaknesses that significantly impact its overall contribution. First and foremost, the experimental validation is severely limited by the choice of model and task. The authors employ a shallow multi-layer perceptron (MLP) with only two hidden layers (64 and 32 units) for a text classification task on the AG News dataset. As I've verified, this dataset is one that large language models (LLMs) are already known to perform well on. The use of such a simple model for a task that LLMs are designed for undermines the paper's claim of enhancing creative diversity in LLMs. The paper states, "Our experiments utilized a shallow multi-layer perceptron (MLP) with two hidden layers, selected for its balance between simplicity and capacity for rapid experimentation." This justification is insufficient given the paper's focus on LLMs. This choice raises serious concerns about the generalizability of the findings to more complex LLMs and tasks where creative diversity is more critical, such as storytelling and content creation. The paper's motivation emphasizes the need for creative diversity in LLMs for open-ended tasks, stating, 'The rapid development of Large Language Models (LLMs) has significantly influenced various sectors... However, the creative potential of LLMs, vital for applications like storytelling and content creation, is hindered by a bias towards generating high-probability sequences, leading to predictable and less original outputs.' The experimental setup directly contradicts this motivation. The second major weakness is the lack of a detailed explanation of the transformation process for generating structured seed variations. The paper introduces the concept of structured seed-conditioning and mentions applying a transformation T to the seed, but the specifics of this transformation are not provided. The paper states, 'Given a seed s , we compute a set of structured variations $\{S_1, S_2, \dots, S_n\}$ by applying a transformation T that strategically alters the seed structure to introduce variability in initial conditions.' However, the description remains at a high level, and the parameters governing the transformation (θ_i) are not elaborated upon. This lack of detail makes it difficult to understand how the structured seed variations are generated and, more importantly, to replicate the experiments. This is a critical omission, as the transformation process is central to the proposed method. The third significant weakness is the insufficient detail regarding the qualitative human assessment. While the paper mentions incorporating qualitative human assessments into the hybrid metric, it lacks specifics about the methodology. The paper states, 'Qualitative Human Assessment, incorporating subjective human judgments of creativity.' However, it does not provide details on the number of evaluators, the instructions given to them, the rubric used, or how inter-rater reliability was ensured. This lack of methodological detail makes it difficult to assess the reliability and validity of the qualitative evaluation. The paper also lacks a clear explanation of how the different components of the hybrid metric are combined into a single score. The paper states, 'We propose a hybrid metric D , integrating: Entropy of the output distribution, Novelty Score, Qualitative Human Assessment.' However, it does not specify the weights or method used to combine these components. This lack of clarity makes it difficult to interpret the results and compare them across different runs. Finally, while the paper cites related work on seed-conditioning, it does not adequately differentiate its approach from existing methods. The paper states, 'Previous research has explored various applications of LLMs but has not deeply examined the optimization of creativity through seed-conditioning.' However, it does not provide a detailed comparison of its method with specific existing seed-conditioning techniques, making it difficult to assess the novelty and contribution of the proposed approach. These weaknesses, particularly the inadequate experimental setup and the lack of methodological details, significantly undermine the paper's claims and limit its overall impact. The lack of generalizability, the missing details on the transformation process, the insufficient information on qualitative evaluation, the unclear combination of metric components, and the inadequate differentiation from prior work all contribute to a less convincing and less impactful study.

Suggestions

To address the identified weaknesses, I recommend several concrete improvements. First and foremost, the experimental validation needs to be significantly strengthened. The authors should conduct experiments using actual large language models on tasks that are more relevant to creative content generation, such as story writing, poetry generation, or creative copywriting. This would provide a more compelling demonstration of the framework's effectiveness in enhancing creative diversity. The use of a shallow MLP on a standard classification dataset is not sufficient to support the paper's claims about LLMs. The authors should also consider using a more diverse set of evaluation metrics that are specifically designed for creative tasks, such as measures of originality, fluency, and coherence. This would provide a more comprehensive assessment of the creative quality of the generated text. Second, the paper needs to provide a detailed explanation of the transformation process for generating structured seed variations. This should include a clear description of the mathematical or algorithmic steps involved in applying the transformation T to the seed input. The authors should also specify the parameters governing the transformation (θ_i) and explain how these parameters are optimized. This level of detail is crucial for understanding and replicating the proposed method. Third, the paper needs to provide a comprehensive description of the qualitative human assessment methodology. This should include details on the number of evaluators, the instructions given to them, the rubric used, and how inter-rater reliability was ensured. The authors should also consider using a more structured approach to qualitative evaluation, such as using a Likert scale to rate different aspects of creativity. This would provide a more reliable and valid assessment of the creative quality of the generated text. Fourth, the paper needs to clearly specify how the different components of the hybrid metric are combined into a single score. This should include a detailed explanation of the weights or method used to combine the entropy, novelty scores, and qualitative human assessments. The authors should also provide a justification for the chosen combination method. This would make the results more interpretable and comparable. Finally, the paper needs to provide a more detailed comparison of its approach with existing seed-conditioning techniques. This should include a discussion of the similarities and differences between the proposed method and specific existing methods, as well as a justification for why the proposed approach is novel and more effective. This would help to clarify the contribution of the paper and its place in the existing literature. By addressing these weaknesses, the authors can significantly strengthen the paper and make a more compelling contribution to the field of LLM research.

? Questions

Several key questions arise from my analysis of the paper. First, how would the proposed structured seed-conditioning framework perform on more complex and diverse datasets that are more representative of real-world creative tasks? The current experiments are limited to the AG News dataset, which is a relatively simple text classification dataset. It would be valuable to see how the framework performs on datasets that are more challenging and more relevant to creative content generation. Second, how does the proposed method compare to other techniques for enhancing creative diversity in LLMs, such as prompt engineering or fine-tuning? The paper does not provide a detailed comparison with these alternative approaches. It would be valuable to understand the strengths and weaknesses of the proposed method relative to these other techniques. Third, what is the computational cost of the proposed method, and how does it scale with the size of the LLM and the complexity of the task? The paper does not provide a detailed analysis of the computational cost. It would be valuable to understand the practical implications of using the proposed method in real-world applications. Fourth, how sensitive is the proposed method to the choice of parameters governing the transformation process? The paper does not provide a detailed analysis of the parameter sensitivity. It would be valuable to understand how the performance of the

method is affected by different parameter settings. Fifth, how can the qualitative human assessment be made more reliable and consistent? The paper does not provide sufficient details about the methodology used for qualitative evaluation. It would be valuable to explore different approaches for improving the reliability and validity of qualitative assessments of creativity. Finally, what are the limitations of the proposed hybrid metric for evaluating creativity? The paper does not provide a detailed discussion of the limitations of the metric. It would be valuable to understand the potential biases and shortcomings of the metric and how they might affect the results. Addressing these questions would provide a more comprehensive understanding of the proposed method and its potential for enhancing creative diversity in LLMs.

 Scores

Soundness: 1.33

Presentation: 1.67

Contribution: 1.33

Rating: 2.33

Keywords

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